Mesh Web Protocol Documentation

A technical description of Mesh Web

github.com/7xy95/MeshWeb
7xy.dev@gmail.com

Last updated on 25/10/2026.

1. Introduction

Domain registration has become a paid subscription on the public DNS through a third party registrar. While the system works well enough for large firms and organisations, it fails ordinary people who would like a short and clear domain name without paying a lot of money. To fix this, Mesh Web allows anyone to register any domain ending with .mesh by using computational power. While there do exist systems in which domain registration is on a blockchain, it is tied to a specific cryptocurrency and requires money to register, which Mesh Web avoids.

2. Registration

Registration is the process where a user creates a valid claim to a .mesh domain. As there is no central registrar, each domain is stored on all nodes as:

[domainName, publicKey, nonce, endpoint, sequenceNumber, signature]

The public key and the signature use ED25519 public key cryptography, allowing the owner to prove the ownership of the private key without revealing it. The signature is taken of the domain, endpoint and sequence number, meaning no one can change the endpoint without mining if the private key is not known as that would make the signature invalid. The sequence number is a unique incrementing number which must increment by 1, so that it becomes clear which endpoint is the most recent one.

To register a domain, it has to be mined, which is the process of guessing different nonces until SHA-256 of (domainName + publicKey + nonce) produces a small enough result. To mine an unregistered domain, a difficulty of at least 31 bits (each bit makes finding a valid hash 2x harder) has to be reached, which is on average 2^31 attempts (about 2.147 billion hashes). If however the domain is already registered, it can still be mined by finding a rarer nonce which produces a hash below the required target. As a result, domain ownership is not permanent while allowing anyone to try and claim any domain with the only limiting factor being computational power and time to mine it.

3. Ownership

All registered domains are not permanent as anyone can challenge an existing domain by finding a nonce which produces a lower hash. While a domain is owned, the endpoint can be changed without needing to re-mine as it only requires incrementing the sequence number and creating a new signature to prove ownership of the domain. The ability to own any already registered domain by finding a better nonce allows anyone to own any domain desired, with the only limitation being computational power and time.

Important domains will face higher competition, and so with each successful takeover the difficulty to reclaim it will continue to grow. As a result, any takeover raises the difficulty by a

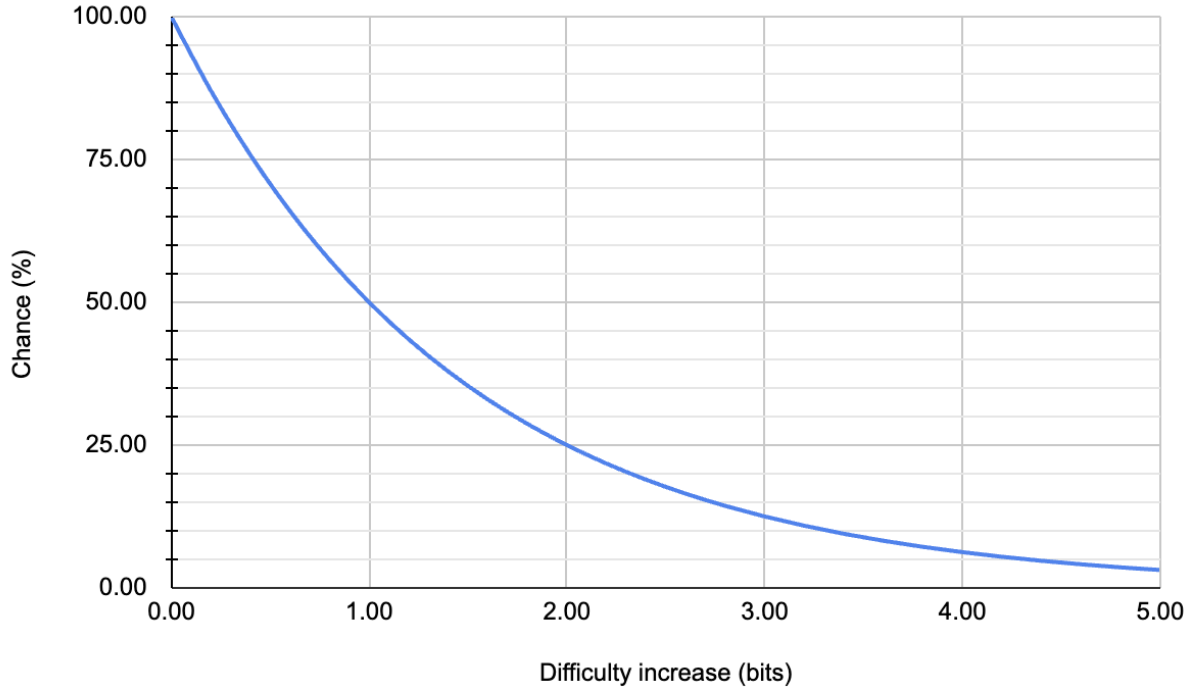


Figure 3a, difficulty increase after taking over a domain

median of 1 bit (2x harder) and a mean of approximately 1.44 bits (2.72x harder). This naturally prevents popular domain names from constantly switching ownership as difficulty continues to increase.

However, mining uses electricity, which can introduce a real world cost into obtaining or overtaking a domain. The expected electricity cost, C, to overtake a domain is on average:

$$C = \frac{2^D PE}{3.6 \times 10^6 R}$$

Where:
D is difficulty in bits
P is power consumption in Watts
E is electricity price per kWh
R is the hashrate

Let's say a domain has a difficulty of 45 bits, and as of 2026 a typical laptop/computer consumes about 100W and can achieve around 500 million hashes per second. In Spain, electricity is around 0.25€ per kWh, as a result, overtaking the domain would cost on average 0.49€. However, it would

take on average 19.5 hours. As a result, the monetary cost of mining can remain relatively low even at higher difficulties, while the time required to complete the process may become a significant practical limitation.

4. Resolution

A .mesh domain is mapped directly to the endpoint, preserving all parameters. Suppose example.mesh maps to example.net, when a user enters example.mesh?param=value Mesh Web looks up the corresponding endpoint and finds it to be example.net, so the user sees example.mesh?param=value as the URL while the browser opens example.net?param=value. This allows mapping non-static pages to a .mesh domain. This allows anyone to convert an IP address which may be hard to memorise to a memorable domain.

5. Network

As any valid domains are accepted, even if most of the nodes in the network are not following the protocol, the network or honest nodes will not break as it doesn't require the majority of the network to follow the protocol. All honest nodes follow the following rules:

1. New valid domains are broadcast to all nodes
2. Any valid domain takeovers or endpoint changes are also broadcast to all nodes
3. New nodes are broadcast to all known nodes
4. Invalid domains or already known nodes are not broadcast as other nodes will reject it anyway
5. Nodes can share existing domain records upon request
6. A new domain which already has an owner is only considered valid if it has a lower hash

During broadcast, each node at random selects 15 nodes from its known node list and shares the data with them.

As honest nodes continue to operate, they will request every 30s some domain names at random to sync in case some information was lost. This will lead each node to converge towards equal and perfect information. As a result, if only a percentage of nodes knows about a new domain registration or takeover, over time this information will spread as nodes continue to sync, which doesn't require the majority of nodes to know the domain for it to still spread across the network. Let x% be the percentage of nodes who know about a domain registration. Assuming all nodes sync every 30s each node requests a random node for 10% of its known domains, then after 30s we would expect x% to become:

$$(1.1x - 0.001x^2)\%$$

Lets say due to an attacker only 20% of nodes receive a domain update, then we would expect:

$$21.60\% \text{ after 30 seconds}$$
$$39.97\% \text{ after 5 minutes}$$
$$99.15\% \text{ after 30 minutes}$$
$$99.99\% \text{ after 50 minutes}$$

As a result, it can be seen that over time even a poorly known domain gets shared across the network, as long as nodes can continue communicating.

6. Limitations

While Mesh Web allows anyone to claim any domain, it also places all domains at risk of being taken over, which might not be favourable for large firms or corporations which want certainty and stability, as even though a lot of mining can be done to raise the difficulty by a lot, an attacker still has a non-zero chance to take over the domain.

If the private key or seed is lost, it becomes impossible for the owner of the domain to change the endpoint, with no ability to recover the private key other than to mine the domain again onto a new private key.

While the mapping of domains to endpoints is controlled, the endpoint is not which exposes the user to privacy and security risks as the endpoint may be malicious.